

The space $H_0^1(\Omega)$, equipped with the norm $\|\cdot\|_{1,\Omega}$, is a Hilbert space. Since $F \in H^{-1}(\Omega)$, it is a bounded linear functional on this space. The assumed boundedness and coercivity of a therefore allow the Lax–Milgram lemma to be applied, giving a unique $y \in H_0^1(\Omega)$ such that $a(y, v) = F(v)$ for every $v \in H_0^1(\Omega)$.

The finite-dimensional subspace V_h^0 is closed in $H_0^1(\Omega)$ and hence is itself a Hilbert space with the inherited norm. The restrictions of a and F to V_h^0 retain the same boundedness and coercivity properties. A second application of the Lax–Milgram lemma gives a unique $y_h \in V_h^0$ satisfying the discrete problem.

Subtracting the two variational equations on V_h^0 gives the Galerkin orthogonality relation $a(y - y_h, w_h) = 0$ for every $w_h \in V_h^0$. Let $e = y - y_h$ and take any $v_h \in V_h^0$. Because $v_h - y_h \in V_h^0$, bilinearity and Galerkin orthogonality imply $a(e, e) = a(e, y - v_h)$. Coercivity and boundedness thus yield

$$c\|e\|_{1,\Omega}^2 \leq a(e, e) = a(e, y - v_h) \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}.$$

If $e = 0$, the desired estimate is immediate; otherwise, division by $c\|e\|_{1,\Omega}$ gives $\|y - y_h\|_{1,\Omega} \leq (C/c)\|y - v_h\|_{1,\Omega}$. Taking the infimum over $v_h \in V_h^0$ proves Céa's estimate.